

# Anesthesia for Hepatic Disease – MCQ

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**Source:** Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Hepatic Disease

**50 Questions • 30 Minutes**

**Instructions:**

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

# Questions

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**Q1. The liver receives what percentage of total cardiac output?**

- A. 10%
- B. 25%
- C. 40%
- D. 50%

**Ans: B**

**Q2. What percentage of hepatic blood supply comes from the portal vein?**

- A. 25%
- B. 50%
- C. 75%
- D. 90%

**Ans: C**

**Q3. Child-Pugh classification assesses liver function based on all EXCEPT:**

- A. Serum bilirubin
- B. Serum albumin
- C. Serum creatinine
- D. Prothrombin time/INR

**Ans: C**

**Q4. Child-Pugh Class C (score 10–15) is associated with perioperative mortality of approximately:**

- A. 5%
- B. 10–15%
- C. 50% or higher
- D. Less than 2%

**Ans: C**

**Q5. Which coagulation factor has the shortest half-life and is first to decrease in liver disease?**

- A. Factor II
- B. Factor V
- C. Factor VII
- D. Factor X

**Ans: C**

**Q6. The most sensitive preoperative liver function test for synthetic function is:**

- A. AST/ALT
- B. Serum albumin
- C. Prothrombin time (PT/INR)
- D. Alkaline phosphatase

**Ans: C**

**Q7. Serum albumin less than what value indicates severe liver disease and increased surgical risk?**

- A. 4.0 g/dL
- B. 3.5 g/dL
- C. 3.0 g/dL
- D. 2.5 g/dL

**Ans: C**

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| <p><b>Q8. In liver disease, which of the following is TRUE about drug protein binding?</b></p> <ul style="list-style-type: none"> <li>A. Increased binding due to elevated albumin</li> <li>B. Decreased binding leading to higher free drug fraction</li> <li>C. No change in drug binding</li> <li>D. Increased binding due to elevated globulin</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q9. Preoperative correction of coagulopathy in liver disease includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Fresh Frozen Plasma (FFP)</li> <li>B. Vitamin K injection</li> <li>C. Platelet transfusion if count &lt;50,000</li> <li>D. Heparin infusion</li> </ul>                                                              | <p><b>Ans: D</b></p> |
| <p><b>Q10. MELD score is calculated using which three parameters?</b></p> <ul style="list-style-type: none"> <li>A. Bilirubin, Albumin, PT</li> <li>B. Bilirubin, Creatinine, INR</li> <li>C. Bilirubin, Ascites, Encephalopathy</li> <li>D. Creatinine, Albumin, PT</li> </ul>                                                                               | <p><b>Ans: B</b></p> |
| <p><b>Q11. In cirrhotic patients, preoperative ascites indicates:</b></p> <ul style="list-style-type: none"> <li>A. Good prognosis for surgery</li> <li>B. Portal hypertension and increased surgical risk</li> <li>C. No impact on anesthetic management</li> <li>D. Need for immediate splenectomy</li> </ul>                                               | <p><b>Ans: B</b></p> |
| <p><b>Q12. Hepatic encephalopathy in the preoperative period is graded from:</b></p> <ul style="list-style-type: none"> <li>A. Grade I to III</li> <li>B. Grade I to IV</li> <li>C. Grade A to C</li> <li>D. Stage 1 to 3</li> </ul>                                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q13. Preoperative optimization of hepatic disease patient includes all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Correction of electrolyte imbalances</li> <li>B. Treatment of encephalopathy with lactulose</li> <li>C. Drainage of tense ascites</li> <li>D. Starting high-dose NSAIDs for pain</li> </ul>                            | <p><b>Ans: D</b></p> |
| <p><b>Q14. Hyperdynamic circulation in cirrhosis is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. High SVR, low cardiac output</li> <li>B. Low SVR, high cardiac output, low MAP</li> <li>C. Normal SVR, normal cardiac output</li> <li>D. High SVR, high cardiac output</li> </ul>                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q15. Which volatile anesthetic agent is most preferred in liver disease?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Isoflurane</li> <li>C. Enflurane</li> <li>D. Desflurane</li> </ul>                                                                                                                                     | <p><b>Ans: B</b></p> |

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| <p><b>Q16. Halothane is avoided in liver disease primarily because:</b></p> <ul style="list-style-type: none"> <li>A. It causes renal failure</li> <li>B. It reduces hepatic blood flow and can cause halothane hepatitis</li> <li>C. It causes seizures</li> <li>D. It increases portal pressure</li> </ul>                             | <p><b>Ans: B</b></p> |
| <p><b>Q17. Which muscle relaxant is the agent of choice in liver disease?</b></p> <ul style="list-style-type: none"> <li>A. Pancuronium</li> <li>B. Vecuronium</li> <li>C. Atracurium/Cisatracurium</li> <li>D. Rocuronium</li> </ul>                                                                                                    | <p><b>Ans: C</b></p> |
| <p><b>Q18. Succinylcholine duration may be prolonged in liver disease because:</b></p> <ul style="list-style-type: none"> <li>A. Increased hepatic metabolism</li> <li>B. Reduced plasma cholinesterase (pseudocholinesterase) synthesis</li> <li>C. Increased renal excretion</li> <li>D. Enhanced neuromuscular sensitivity</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q19. Which opioid is preferred in liver disease due to minimal hepatic metabolism?</b></p> <ul style="list-style-type: none"> <li>A. Morphine</li> <li>B. Fentanyl</li> <li>C. Remifentanyl</li> <li>D. Codeine</li> </ul>                                                                                                         | <p><b>Ans: C</b></p> |
| <p><b>Q20. Induction agent of choice in liver disease is:</b></p> <ul style="list-style-type: none"> <li>A. Thiopentone in full dose</li> <li>B. Propofol in reduced dose</li> <li>C. Ketamine</li> <li>D. Etomidate in full dose</li> </ul>                                                                                             | <p><b>Ans: B</b></p> |
| <p><b>Q21. Intraoperative monitoring in liver disease should include:</b></p> <ul style="list-style-type: none"> <li>A. Only standard ASA monitoring</li> <li>B. Invasive arterial BP, CVP, urine output, temperature, and coagulation</li> <li>C. Only pulse oximetry and ECG</li> <li>D. Only BIS monitoring</li> </ul>                | <p><b>Ans: B</b></p> |
| <p><b>Q22. Regional anesthesia in liver disease is relatively contraindicated when:</b></p> <ul style="list-style-type: none"> <li>A. Albumin is normal</li> <li>B. INR &gt;1.5 or platelets &lt;80,000</li> <li>C. Patient has mild ascites</li> <li>D. Bilirubin is mildly elevated</li> </ul>                                         | <p><b>Ans: B</b></p> |
| <p><b>Q23. Intraoperative hypoglycemia in liver disease occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Excess insulin production</li> <li>B. Impaired hepatic glycogenolysis and gluconeogenesis</li> <li>C. Increased peripheral glucose utilization</li> <li>D. Renal glucose wasting</li> </ul>                    | <p><b>Ans: B</b></p> |

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| <p><b>Q24. Intraoperative fluid management in liver disease should avoid:</b></p> <ul style="list-style-type: none"> <li>A. Normal saline</li> <li>B. Lactated Ringer's solution in severe liver disease</li> <li>C. Albumin</li> <li>D. Packed red blood cells</li> </ul>                                                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q25. Which factor most decreases hepatic blood flow during anesthesia?</b></p> <ul style="list-style-type: none"> <li>A. IPPV with high airway pressures</li> <li>B. Adequate hydration</li> <li>C. Supine position</li> <li>D. Use of isoflurane</li> </ul>                                                                                                                                                | <p><b>Ans: A</b></p> |
| <p><b>Q26. Factors that reduce hepatic blood flow intraoperatively include all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Hypotension</li> <li>B. High sympathetic stimulation</li> <li>C. IPPV</li> <li>D. Adequate intravascular volume</li> </ul>                                                                                                                                               | <p><b>Ans: D</b></p> |
| <p><b>Q27. Temperature regulation in liver disease is impaired due to:</b></p> <ul style="list-style-type: none"> <li>A. Increased basal metabolic rate</li> <li>B. Reduced hepatic heat production and vasodilation</li> <li>C. Increased shivering threshold</li> <li>D. Hypermetabolic state</li> </ul>                                                                                                        | <p><b>Ans: B</b></p> |
| <p><b>Q28. Benzodiazepines in liver disease have:</b></p> <ul style="list-style-type: none"> <li>A. Shorter duration of action</li> <li>B. Prolonged duration due to reduced metabolism and increased free fraction</li> <li>C. No change in pharmacokinetics</li> <li>D. Increased hepatic clearance</li> </ul>                                                                                                  | <p><b>Ans: B</b></p> |
| <p><b>Q29. Which benzodiazepine is relatively safer in liver disease due to glucuronidation (Phase II) metabolism?</b></p> <ul style="list-style-type: none"> <li>A. Diazepam</li> <li>B. Midazolam</li> <li>C. Oxazepam/Lorazepam</li> <li>D. Chlordiazepoxide</li> </ul>                                                                                                                                        | <p><b>Ans: C</b></p> |
| <p><b>Q30. The hepatic artery buffer response refers to:</b></p> <ul style="list-style-type: none"> <li>A. Increase in portal vein flow when hepatic artery flow decreases</li> <li>B. Compensatory increase in hepatic artery flow when portal venous flow decreases</li> <li>C. Shunting of blood through portosystemic collaterals</li> <li>D. Decrease in both hepatic artery and portal vein flow</li> </ul> | <p><b>Ans: B</b></p> |

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| <p><b>Q31. Intraoperative management of ascites includes:</b></p> <ul style="list-style-type: none"> <li>A. Ignoring it during surgery</li> <li>B. Rapid paracentesis of all fluid at induction</li> <li>C. Careful drainage with albumin replacement to prevent cardiovascular collapse</li> <li>D. Using high-dose diuretics during surgery</li> </ul>                         | <p><b>Ans: C</b></p> |
| <p><b>Q32. Hepatorenal syndrome is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Structural renal damage</li> <li>B. Functional renal failure due to splanchnic vasodilation and renal vasoconstriction</li> <li>C. Obstructive uropathy</li> <li>D. Glomerulonephritis</li> </ul>                                                                         | <p><b>Ans: B</b></p> |
| <p><b>Q33. Obstructive jaundice (biliary obstruction) is characterized by:</b></p> <ul style="list-style-type: none"> <li>A. Elevated unconjugated bilirubin only</li> <li>B. Elevated conjugated bilirubin, raised ALP, pale stools, dark urine</li> <li>C. Normal bilirubin with elevated transaminases</li> <li>D. Elevated unconjugated bilirubin with normal ALP</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q34. In obstructive jaundice, Vitamin K deficiency occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Increased renal excretion of Vitamin K</li> <li>B. Lack of bile salts prevents fat-soluble vitamin absorption from gut</li> <li>C. Increased hepatic metabolism of Vitamin K</li> <li>D. Dietary deficiency</li> </ul>                                | <p><b>Ans: B</b></p> |
| <p><b>Q35. In obstructive jaundice, coagulopathy from Vitamin K deficiency responds to parenteral Vitamin K within:</b></p> <ul style="list-style-type: none"> <li>A. 1-2 hours</li> <li>B. 6-12 hours</li> <li>C. 24-48 hours</li> <li>D. 5-7 days</li> </ul>                                                                                                                   | <p><b>Ans: C</b></p> |
| <p><b>Q36. The major renal risk in obstructive jaundice is:</b></p> <ul style="list-style-type: none"> <li>A. Renal calculi</li> <li>B. Acute tubular necrosis (hepatorenal syndrome/bile nephropathy)</li> <li>C. Renal artery stenosis</li> <li>D. Pyelonephritis</li> </ul>                                                                                                   | <p><b>Ans: B</b></p> |
| <p><b>Q37. Renal protection in obstructive jaundice includes:</b></p> <ul style="list-style-type: none"> <li>A. Fluid restriction</li> <li>B. Adequate preoperative hydration and maintaining urine output</li> <li>C. Use of nephrotoxic antibiotics</li> <li>D. High-dose loop diuretics</li> </ul>                                                                            | <p><b>Ans: B</b></p> |

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| <p><b>Q38. Endotoxemia in obstructive jaundice occurs because:</b></p> <ul style="list-style-type: none"> <li>A. Excess bile production</li> <li>B. Absence of bile in gut allows bacterial overgrowth and endotoxin absorption</li> <li>C. Direct liver cell necrosis</li> <li>D. Renal failure</li> </ul>                                                                    | <p><b>Ans: B</b></p> |
| <p><b>Q39. Preoperative biliary drainage (PTBD/stenting) before surgery in obstructive jaundice is considered when:</b></p> <ul style="list-style-type: none"> <li>A. Bilirubin is mildly elevated</li> <li>B. Bilirubin &gt;10 mg/dL with planned major surgery, to reduce complications</li> <li>C. Patient is asymptomatic</li> <li>D. Only for cosmetic reasons</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q40. Anesthetic technique for patients with biliary obstruction preferably avoids:</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Morphine and drugs causing sphincter of Oddi spasm</li> <li>C. Fentanyl</li> <li>D. Atracurium</li> </ul>                                                                                                      | <p><b>Ans: B</b></p> |
| <p><b>Q41. Which drug can relieve morphine-induced sphincter of Oddi spasm?</b></p> <ul style="list-style-type: none"> <li>A. Atropine</li> <li>B. Glucagon</li> <li>C. Adrenaline</li> <li>D. Propranolol</li> </ul>                                                                                                                                                          | <p><b>Ans: B</b></p> |
| <p><b>Q42. Pruritus in obstructive jaundice is caused by:</b></p> <ul style="list-style-type: none"> <li>A. Elevated bilirubin directly</li> <li>B. Deposition of bile salts in the skin</li> <li>C. Bacterial infection</li> <li>D. Vitamin K deficiency</li> </ul>                                                                                                           | <p><b>Ans: B</b></p> |
| <p><b>Q43. Wound healing in obstructive jaundice is impaired due to:</b></p> <ul style="list-style-type: none"> <li>A. Excess Vitamin A</li> <li>B. Malnutrition, Vitamin K deficiency, and reduced immunity</li> <li>C. Excess collagen production</li> <li>D. Increased blood supply to wound</li> </ul>                                                                     | <p><b>Ans: B</b></p> |
| <p><b>Q44. Bradycardia is more common in jaundiced patients because:</b></p> <ul style="list-style-type: none"> <li>A. Pain</li> <li>B. Bile salts sensitize the myocardium to vagal stimulation</li> <li>C. Hypovolemia</li> <li>D. Hypoglycemia</li> </ul>                                                                                                                   | <p><b>Ans: B</b></p> |

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| <p><b>Q45. In obstructive jaundice, which antibiotic prophylaxis is important preoperatively?</b></p> <ul style="list-style-type: none"> <li>A. Not needed</li> <li>B. Broad-spectrum antibiotics covering gram-negative and anaerobic organisms</li> <li>C. Only antifungals</li> <li>D. Only gram-positive coverage</li> </ul>                      | <p><b>Ans: B</b></p> |
| <p><b>Q46. Vecuronium and rocuronium should be used cautiously in liver disease because:</b></p> <ul style="list-style-type: none"> <li>A. They cause histamine release</li> <li>B. They undergo significant hepatic metabolism and biliary excretion</li> <li>C. They worsen hepatic encephalopathy</li> <li>D. They cause liver necrosis</li> </ul> | <p><b>Ans: B</b></p> |
| <p><b>Q47. Postoperative jaundice in previously healthy patient may be caused by all EXCEPT:</b></p> <ul style="list-style-type: none"> <li>A. Blood transfusion (hemolysis)</li> <li>B. Halothane hepatitis</li> <li>C. Hypotension-related hepatic ischemia</li> <li>D. Use of atracurium</li> </ul>                                                | <p><b>Ans: D</b></p> |
| <p><b>Q48. In biliary surgery, intraoperative cholangiography may produce false positive results if which drug has been given?</b></p> <ul style="list-style-type: none"> <li>A. Propofol</li> <li>B. Morphine (sphincter of Oddi spasm mimics stone)</li> <li>C. Atracurium</li> <li>D. Sevoflurane</li> </ul>                                       | <p><b>Ans: B</b></p> |
| <p><b>Q49. Mannitol is sometimes used in obstructive jaundice surgery to:</b></p> <ul style="list-style-type: none"> <li>A. Reduce intracranial pressure</li> <li>B. Maintain renal perfusion and urine output (renal protection)</li> <li>C. Reduce portal pressure</li> <li>D. Treat hyperkalemia</li> </ul>                                        | <p><b>Ans: B</b></p> |
| <p><b>Q50. Which is the safest volatile agent regarding hepatic blood flow preservation?</b></p> <ul style="list-style-type: none"> <li>A. Halothane</li> <li>B. Enflurane</li> <li>C. Isoflurane/Sevoflurane</li> <li>D. Desflurane at high MAC</li> </ul>                                                                                           | <p><b>Ans: C</b></p> |



# Answer Key & Explanations

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## Q1. Answer: B — 25%

Correct Answer: B. 25% — The liver receives approximately 25% of cardiac output (~1500 mL/min). Of this, 75% comes from the portal vein and 25% from the hepatic artery.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q2. Answer: C — 75%

Correct Answer: C. 75% — The portal vein provides 75% of hepatic blood flow but only 50% of oxygen supply. The hepatic artery provides 25% of blood flow but 50% of oxygen supply.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q3. Answer: C — Serum creatinine

Correct Answer: C. Serum creatinine — Child-Pugh score uses 5 parameters: Bilirubin, Albumin, Prothrombin time (INR), Ascites, and Encephalopathy grade. Creatinine is part of the MELD score, not Child-Pugh.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q4. Answer: C — 50% or higher

Correct Answer: C. 50% or higher — Child-Pugh C carries perioperative mortality >50% for major abdominal surgery. Class A: ~10%, Class B: ~30%, Class C: >50%.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q5. Answer: C — Factor VII

Correct Answer: C. Factor VII — Factor VII has the shortest half-life (~6 hours) among coagulation factors and is the first to decline in liver disease. Its level is reflected by PT/INR.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q6. Answer: C — Prothrombin time (PT/INR)

Correct Answer: C. Prothrombin time (PT/INR) — PT is the most sensitive indicator of hepatic synthetic function due to the short half-life of Factor VII. Albumin reflects chronic liver function (half-life 21 days).

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q7. Answer: C — 3.0 g/dL

Correct Answer: C. 3.0 g/dL — Albumin <3.0 g/dL indicates severe hepatic dysfunction. Low albumin causes reduced protein binding of drugs, increased free drug levels, and peripheral edema/ascites.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q8. Answer: B — Decreased binding leading to higher free drug fraction

Correct Answer: B. Decreased binding leading to higher free drug fraction — Hypoalbuminemia in liver disease results in reduced protein binding of drugs (thiopentone, diazepam), causing increased free drug levels and exaggerated clinical effects.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

## Q9. Answer: D — Heparin infusion

Correct Answer: D. Heparin infusion — Heparin would worsen coagulopathy. Treatment includes Vitamin K (if cholestatic), FFP for active bleeding, cryoprecipitate for low fibrinogen, and platelet transfusion if <50,000/ $\mu$ L.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q10. Answer: B — Bilirubin, Creatinine, INR**

Correct Answer: B. Bilirubin, Creatinine, INR — MELD (Model for End-stage Liver Disease) uses serum bilirubin, creatinine, and INR. MELD >15 indicates significant surgical risk.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q11. Answer: B — Portal hypertension and increased surgical risk**

Correct Answer: B. Portal hypertension and increased surgical risk — Ascites reflects portal hypertension and poor hepatic function. It impairs diaphragmatic excursion, increases aspiration risk, and is associated with higher perioperative mortality.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q12. Answer: B — Grade I to IV**

Correct Answer: B. Grade I to IV — Grade I: altered mood/sleep; Grade II: drowsiness, inappropriate behavior; Grade III: stupor, confusion; Grade IV: coma. Surgery should be deferred in Grade III-IV.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q13. Answer: D — Starting high-dose NSAIDs for pain**

Correct Answer: D. Starting high-dose NSAIDs for pain — NSAIDs impair renal function (hepatorenal risk), reduce platelet function, and may precipitate GI bleeding. They should be avoided in liver disease patients.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q14. Answer: B — Low SVR, high cardiac output, low MAP**

Correct Answer: B. Low SVR, high cardiac output, low MAP — Cirrhotic patients have splanchnic vasodilation (due to nitric oxide), reduced SVR, compensatory high cardiac output, and relatively low mean arterial pressure.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q15. Answer: B — Isoflurane**

Correct Answer: B. Isoflurane — Isoflurane maintains hepatic arterial blood flow better than other agents. Halothane reduces hepatic blood flow the most and is associated with halothane hepatitis. Isoflurane and sevoflurane are preferred.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q16. Answer: B — It reduces hepatic blood flow and can cause halothane hepatitis**

Correct Answer: B. It reduces hepatic blood flow and can cause halothane hepatitis — Halothane reduces hepatic arterial flow by 25-30% and is associated with immune-mediated fulminant hepatitis (1:35,000 adults). It is contraindicated in liver disease.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q17. Answer: C — Atracurium/Cisatracurium**

Correct Answer: C. Atracurium/Cisatracurium — Atracurium undergoes Hofmann degradation (organ-independent) and ester hydrolysis. Its metabolism is independent of hepatic and renal function, making it ideal in liver disease.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q18. Answer: B — Reduced plasma cholinesterase (pseudocholinesterase) synthesis**

Correct Answer: B. Reduced plasma cholinesterase (pseudocholinesterase) synthesis — The liver synthesizes plasma cholinesterase. Severe liver disease reduces its production, prolonging succinylcholine block duration.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q19. Answer: C — Remifentanyl**

Correct Answer: C. Remifentanyl — Remifentanyl is metabolized by tissue esterases (organ-independent). Morphine has prolonged action due to reduced glucuronidation. Fentanyl metabolism is also hepatic-dependent.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q20. Answer: B — Propofol in reduced dose**

Correct Answer: B. Propofol in reduced dose — Propofol has both hepatic and extrahepatic metabolism. Dose should be reduced due to hypoalbuminemia (increased free fraction) and reduced hepatic clearance.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q21. Answer: B — Invasive arterial BP, CVP, urine output, temperature, and coagulation**

Correct Answer: B. Invasive arterial BP, CVP, urine output, temperature, and coagulation — Due to coagulopathy, hemodynamic instability, renal vulnerability, and hypothermia risk, comprehensive invasive monitoring is essential.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q22. Answer: B — INR >1.5 or platelets <80,000**

Correct Answer: B. INR >1.5 or platelets <80,000 — Neuraxial techniques carry risk of epidural/spinal hematoma in coagulopathy. Generally contraindicated if INR >1.5, platelets <80,000, or active coagulopathy.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q23. Answer: B — Impaired hepatic glycogenolysis and gluconeogenesis**

Correct Answer: B. Impaired hepatic glycogenolysis and gluconeogenesis — The liver is the primary organ for maintaining blood glucose. In severe disease, depleted glycogen stores and impaired gluconeogenesis cause hypoglycemia.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q24. Answer: B — Lactated Ringer's solution in severe liver disease**

Correct Answer: B. Lactated Ringer's solution in severe liver disease — The diseased liver cannot metabolize lactate to bicarbonate efficiently. Lactate accumulation may worsen acidosis. Normal saline or balanced crystalloids without lactate are preferred.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q25. Answer: A — IPPV with high airway pressures**

Correct Answer: A. IPPV with high airway pressures — Positive pressure ventilation increases intrathoracic pressure, reducing venous return and thereby reducing portal venous flow. High PEEP worsens this effect.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q26. Answer: D — Adequate intravascular volume**

Correct Answer: D. Adequate intravascular volume — Maintaining intravascular volume preserves hepatic perfusion. Hypotension, sympathetic stimulation (surgical stress), IPPV, and vasoconstrictors all reduce hepatic blood flow.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q27. Answer: B — Reduced hepatic heat production and vasodilation**

Correct Answer: B. Reduced hepatic heat production and vasodilation — The liver contributes significantly to heat production. Cirrhotic vasodilation increases heat loss. Active warming and temperature monitoring are essential.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q28. Answer: B — Prolonged duration due to reduced metabolism and increased free fraction**

Correct Answer: B. Prolonged duration due to reduced metabolism and increased free fraction — Hepatic oxidation of benzodiazepines is impaired in liver disease. Combined with hypoalbuminemia, this causes prolonged and exaggerated sedation.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q29. Answer: C — Oxazepam/Lorazepam**

Correct Answer: C. Oxazepam/Lorazepam — Oxazepam and lorazepam undergo Phase II conjugation (glucuronidation) which is relatively preserved in liver disease, unlike Phase I oxidation (diazepam, midazolam).

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q30. Answer: B — Compensatory increase in hepatic artery flow when portal venous flow decreases**

Correct Answer: B. Compensatory increase in hepatic artery flow when portal venous flow decreases — This semi-reciprocal mechanism maintains total hepatic blood flow and oxygen delivery when portal flow is reduced.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q31. Answer: C — Careful drainage with albumin replacement to prevent cardiovascular collapse**

Correct Answer: C. Careful drainage with albumin replacement to prevent cardiovascular collapse — Large volume paracentesis (>5L) requires albumin replacement (6-8g per liter removed) to prevent post-paracentesis circulatory dysfunction.

Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative

**Q32. Answer: B — Functional renal failure due to splanchnic vasodilation and renal vasoconstriction**

Correct Answer: B. Functional renal failure due to splanchnic vasodilation and renal vasoconstriction — Hepatorenal syndrome is functional (kidneys are structurally normal). Splanchnic vasodilation causes underfilling, activating RAAS and causing renal vasoconstriction.

Topic: Preoperative Evaluation | Ref: Ch 21, Page 199, Topic: Preoperative Evaluation

**Q33. Answer: B — Elevated conjugated bilirubin, raised ALP, pale stools, dark urine**

Correct Answer: B. Elevated conjugated bilirubin, raised ALP, pale stools, dark urine — Biliary obstruction causes conjugated hyperbilirubinemia, markedly elevated alkaline phosphatase and GGT, acholic stools, and choluria.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q34. Answer: B — Lack of bile salts prevents fat-soluble vitamin absorption from gut**

Correct Answer: B. Lack of bile salts prevents fat-soluble vitamin absorption from gut — Bile salts are essential for absorption of fat-soluble vitamins (A, D, E, K). Without bile in the gut, Vitamin K malabsorption causes coagulopathy.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q35. Answer: C — 24-48 hours**

Correct Answer: C. 24-48 hours — Parenteral Vitamin K corrects coagulopathy in obstructive jaundice within 24-48 hours. If PT does not correct with Vitamin K, it suggests hepatocellular damage rather than pure obstruction.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q36. Answer: B — Acute tubular necrosis (hepatorenal syndrome/bile nephropathy)**

Correct Answer: B. Acute tubular necrosis (hepatorenal syndrome/bile nephropathy) — Obstructive jaundice predisposes to renal failure due to direct bilirubin toxicity to renal tubules, endotoxemia, and reduced renal perfusion.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q37. Answer: B — Adequate preoperative hydration and maintaining urine output**

Correct Answer: B. Adequate preoperative hydration and maintaining urine output — Preoperative IV fluids, maintaining urine output >0.5-1 mL/kg/hr, and avoiding nephrotoxic drugs are essential for renal protection in jaundiced patients.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q38. Answer: B — Absence of bile in gut allows bacterial overgrowth and endotoxin absorption**

Correct Answer: B. Absence of bile in gut allows bacterial overgrowth and endotoxin absorption — Bile salts normally prevent bacterial overgrowth and endotoxin absorption from the gut. Their absence leads to endotoxemia, predisposing to sepsis and renal failure.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q39. Answer: B — Bilirubin >10 mg/dL with planned major surgery, to reduce complications**

Correct Answer: B. Bilirubin >10 mg/dL with planned major surgery, to reduce complications — Preoperative biliary drainage may reduce renal complications, sepsis, and wound healing problems in severely jaundiced patients undergoing major procedures.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q40. Answer: B — Morphine and drugs causing sphincter of Oddi spasm**

Correct Answer: B. Morphine and drugs causing sphincter of Oddi spasm — Morphine causes sphincter of Oddi spasm (up to 50-fold increase in biliary pressure). Fentanyl has less effect; glucagon or naloxone can reverse spasm if needed.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q41. Answer: B — Glucagon**

Correct Answer: B. Glucagon — Glucagon relaxes the sphincter of Oddi smooth muscle. Naloxone (opioid reversal), nitroglycerin, and volatile anesthetics also help relieve opioid-induced biliary spasm.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q42. Answer: B — Deposition of bile salts in the skin**

Correct Answer: B. Deposition of bile salts in the skin — Retained bile salts deposit in the skin and stimulate nerve endings causing intense pruritus. Cholestyramine (bile salt binder) and rifampin may provide relief.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q43. Answer: B — Malnutrition, Vitamin K deficiency, and reduced immunity**

Correct Answer: B. Malnutrition, Vitamin K deficiency, and reduced immunity — Obstructive jaundice impairs wound healing through protein malnutrition, fat-soluble vitamin deficiency, impaired immune function, and endotoxemia.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q44. Answer: B — Bile salts sensitize the myocardium to vagal stimulation**

Correct Answer: B. Bile salts sensitize the myocardium to vagal stimulation — Elevated bile salts increase cardiac sensitivity to vagal tone and depress myocardial contractility. Atropine should be readily available.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q45. Answer: B — Broad-spectrum antibiotics covering gram-negative and anaerobic organisms**

Correct Answer: B. Broad-spectrum antibiotics covering gram-negative and anaerobic organisms — Cholangitis is common; biliary organisms include E. coli, Klebsiella, Enterococcus, and anaerobes. Prophylaxis with piperacillin-tazobactam or similar is important.

Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction

**Q46. Answer: B — They undergo significant hepatic metabolism and biliary excretion**

Correct Answer: B. They undergo significant hepatic metabolism and biliary excretion — Vecuronium (40-60% biliary excretion) and rocuronium (hepatic uptake) have prolonged duration in liver disease. Atracurium/cisatracurium are preferred alternatives.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

**Q47. Answer: D — Use of atracurium**

Correct Answer: D. Use of atracurium — Atracurium does not affect the liver. Postoperative jaundice can result from blood transfusion hemolysis, drug-induced hepatitis (halothane), hepatic ischemia (hypotension), or sepsis.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*

**Q48. Answer: B — Morphine (sphincter of Oddi spasm mimics stone)**

Correct Answer: B. Morphine (sphincter of Oddi spasm mimics stone) — Morphine-induced sphincter of Oddi spasm can prevent contrast flow into the duodenum, mimicking a common bile duct stone on cholangiography.

*Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction*

**Q49. Answer: B — Maintain renal perfusion and urine output (renal protection)**

Correct Answer: B. Maintain renal perfusion and urine output (renal protection) — Mannitol (osmotic diuretic) helps maintain renal tubular flow and prevent bile pigment-induced tubular damage in jaundiced patients.

*Topic: Anesthesia for Biliary Obstruction | Ref: Ch 21, Page 200, Topic: Anesthesia for Patients with Biliary Obstruction*

**Q50. Answer: C — Isoflurane/Sevoflurane**

Correct Answer: C. Isoflurane/Sevoflurane — Isoflurane and sevoflurane best preserve hepatic arterial buffer response and total hepatic blood flow. Halothane causes the greatest reduction in hepatic blood flow.

*Topic: Intraoperative | Ref: Ch 21, Page 199, Topic: Intraoperative*